

Containerlab

HPE Aruba AOS-CX VSX & Juniper vJunos-switch MC-LAG

Network Virtualization Lab Guide

Nicolas Culetto — Channel Presales, HPE Aruba Networking

github.com/Luconik/netdevops | Avril 2026 / April 2026

Version AOS-CX	AOS-CX 10.16.1020
Version vJunos-switch	25.4R1.12
Containerlab	0.74.3
Plateforme / Platform	Fedora 43 bare metal (i9-10850K / 128GB RAM)
Repo GitHub	github.com/Luconik/netdevops
Guide compte HPE / HPE account guide	github.com/Luconik/hpe-aruba-guides/tree/main/nsp-account

Sommaire / Table of Contents

- 1. Introduction
- 2. Prérequis / Prerequisites
- 3. Lab AOS-CX VSX — HPE Aruba
- 4. Lab Juniper vJunos-switch MC-LAG
- 5. Comparaison VSX vs MC-LAG
- 6. Plugin VS Code Containerlab
- 7. Commandes de référence / Reference Commands

1. Introduction

Objectif / Goal

Ce guide décrit la mise en place de deux labs réseau virtualisés avec **Containerlab** — l'un avec des switches **HPE Aruba AOS-CX** en configuration VSX (Virtual Switching Extension), l'autre avec des switches **Juniper vJunos-switch** en configuration MC-LAG. Ces labs permettent de pratiquer les technologies d'agrégation multi-châssis des deux gammes HPE Networking sans matériel physique.

This guide describes the setup of two virtualized network labs using Containerlab — one with HPE Aruba AOS-CX switches in VSX configuration, the other with Juniper vJunos-switch in MC-LAG configuration. These labs enable practicing multi-chassis aggregation technologies from both HPE Networking product lines without physical hardware.

Containerlab

Containerlab est un outil open-source qui déploie des topologies réseau virtualisées à partir d'un simple fichier YAML (*topology.yml*). Il supporte les images Docker natives et les VMs packagées via **vrnetlab** (QEMU dans Docker). Le plugin VS Code Containerlab offre une visualisation graphique de la topologie et un accès SSH direct par clic droit sur les nodes, compatible avec VS Code Remote SSH.

2. Prérequis / Prerequisites

Infrastructure

Composant	Valeur / Value
OS	Linux bare metal — Fedora 43 / Ubuntu 24.04
CPU	x86_64 avec virtualisation KVM
RAM	16 GB minimum (vJunos ~5 GB par switch)
Containerlab	≥ 0.74.3 — containerlab.dev/install
Docker	Version récente / Recent version
VS Code	Extension Containerlab by srl-labs (optionnel)

Images requises / Required Images

Image Docker	Source	Compte / Account
vrnetlab/vr-aoscx:10.16.1020	networking.hpe.com — AOS-CX Switch Simulator	Oui / Yes *
vrnetlab/juniper_vjunos-switch:25.4R1.12	juniper.net/us/en/dm/vjunos-labs.html	Non / No

* Guide création compte HPE / HPE account creation guide :
github.com/Luconik/hpe-aruba-guides/tree/main/nsp-account

Build des images vrnetlab / Building Images

AOS-CX :

```
tar xf AOS-CX_10_16_1020.ova
cp AOS-CX_10_16_1020.vmdk vrnetlab/aruba/
cd vrnetlab/aruba && make
```

vJunos-switch :

```
cp vJunos-switch-25.4R1.12.qcow2 vrnetlab/juniper/vjunosswitch/
cd vrnetlab/juniper/vjunosswitch && make
# Build ~10 min – image finale ~9 GB / Final image ~9 GB
```

3. Lab AOS-CX VSX — HPE Aruba

Liens / Links

Lien / Link	Interfaces	Rôle / Role
ISL-1	sw1:eth1 ↔ sw2:eth1	Inter-Switch Link (LAG)
ISL-2	sw1:eth2 ↔ sw2:eth2	Inter-Switch Link (LAG)
Keepalive	sw1:eth3 ↔ sw2:eth3	VSX Keepalive

topology.yml

```
name: aoscx-vsx
```

```
topology:
```

```
  nodes:
```

```
    sw1:
```

```
      kind: vr-aoscx
```

```
      image: vrnetlab/vr-aoscx:10.16.1020
```

```
    sw2:
```

```
      kind: vr-aoscx
```

```
      image: vrnetlab/vr-aoscx:10.16.1020
```

```
  links:
```

```
    - endpoints: ["sw1:eth1", "sw2:eth1"] # ISL-1
```

```
    - endpoints: ["sw1:eth2", "sw2:eth2"] # ISL-2
```

```
    - endpoints: ["sw1:eth3", "sw2:eth3"] # VSX Keepalive
```

Déploiement / Deployment

```
sudo containerlab deploy -t topology.yml
```

```
sudo containerlab inspect --all
```

```
# Attendre healthy (~2-3 min) / Wait for healthy state (~2-3 min)
```

```
ssh admin@172.20.20.3 # sw1 — admin/admin
```

```
ssh admin@172.20.20.2 # sw2 — admin/admin
```

EXPLORER

AOSC-CX-LAB [SSH: BAKASTA...]

clab-aoscx-lab

.tls

sw1

sw2

ansible-inventory.yml

authorized_keys

nornir-simple-inventory.yml

topology-data.json

clab-aoscx-vsx

topology.yml

topology.yml

```

1 name: aoscx-vsx
2
3 topology:
4   nodes:
5     sw1:
6       kind: vr-aoscx
7       image: vrnetlab/vr-aoscx:10.16.1020
8     sw2:
9       kind: vr-aoscx
10      image: vrnetlab/vr-aoscx:10.16.1020
11
12 links:
13   - endpoints: ["sw1:eth1", "sw2:eth1"] # ISL
14   - endpoints: ["sw1:eth2", "sw2:eth2"] # keepalive
15

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

bash

```

luconik@bakastation:/mnt/containerlab$ sudo containerlab deploy -t topology.yml
[sudo] password for luconik:
18:26:31 INFO Containerlab started version=0.74.3
18:26:31 INFO Parsing & checking topology file=topology.yml
18:26:31 INFO Creating docker network name=clab IPv4 subnet=172.20.0/24 IPv6 subnet=3fff:172:20:20::/64 MTU=0
18:26:31 INFO Creating lab directory path=/mnt/containerlab/aoscx-lab/clab-aoscx-vsx
18:26:31 INFO Creating container name=sw1
18:26:32 INFO Created link: sw1:eth1 --- sw2:eth1
18:26:32 INFO Created link: sw1:eth2 --- sw2:eth2
18:26:32 INFO Adding host entries path=/etc/hosts
18:26:32 INFO Adding SSH config for nodes path=/etc/ssh/ssh_config.d/clab-aoscx-vsx.conf

```

Name	Kind/Image	State	IPv4/6 Address
clab-aoscx-vsx-sw1	vr-aoscx vrnetlab/vr-aoscx:10.16.1020	running (health: starting)	172.20.20.2 3fff:172:20:20::2
clab-aoscx-vsx-sw2	vr-aoscx vrnetlab/vr-aoscx:10.16.1020	running (health: starting)	172.20.20.3 3fff:172:20:20::3

Déploiement du lab AOS-CX — containers en cours de démarrage AOS-CX lab deployment — containers starting up

CONTAINERLAB EXPLORER

Filter labs, nodes, interfaces

Running Labs (1)

aoscx-vsx (luconik)

Undeployed Local Labs (0)

No items found.

Help & Feedback

Containerlab Documentation

VS Code Extension Documentation

Browse Labs on GitHub (all-labs)

Find more labs tagged with "clab-1..."

Join our Discord server

Download cshargextcap Wireshark ...

topology.yml

```

1 name: aoscx-vsx
2
3 topology:
4   nodes:
5     sw1:
6       kind: vr-aoscx
7       image: vrnetlab/vr-aoscx:10.16.1020
8     sw2:
9       kind: vr-aoscx
10      image: vrnetlab/vr-aoscx:10.16.1020
11
12 links:
13   - endpoints: ["sw1:eth1", "sw2:eth1"] # ISL-1
14   - endpoints: ["sw1:eth2", "sw2:eth2"] # ISL-2
15   - endpoints: ["sw1:eth3", "sw2:eth3"] # VSX Keepalive

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

luconik@bakastation:/mnt/containerlab/aoscx-lab\$

```

luconik@bakastation:/mnt/containerlab/aoscx-lab$ sudo containerlab inspect --all

```

Topology	Lab Name	Name	Kind/Image	State	IPv4/6 Address
topology.yml	aoscx-vsx	clab-aoscx-vsx-sw1	vr-aoscx vrnetlab/vr-aoscx:10.16.1020	running (healthy)	172.20.20.3 3fff:172:20:20::3
		clab-aoscx-vsx-sw2	vr-aoscx vrnetlab/vr-aoscx:10.16.1020	running (healthy)	172.20.20.2 3fff:172:20:20::2

Lab déployé — topology graphique VS Code + 2 switches healthy Lab deployed — VS Code graphical topology + 2 healthy switches

Configuration VSX — SW1 (Primary)

configure terminal

```
interface lag 1
  no shutdown
  no routing
  vlan trunk allowed all
  lacp mode active
```

```
interface 1/1/1
  no shutdown
  lag 1
```

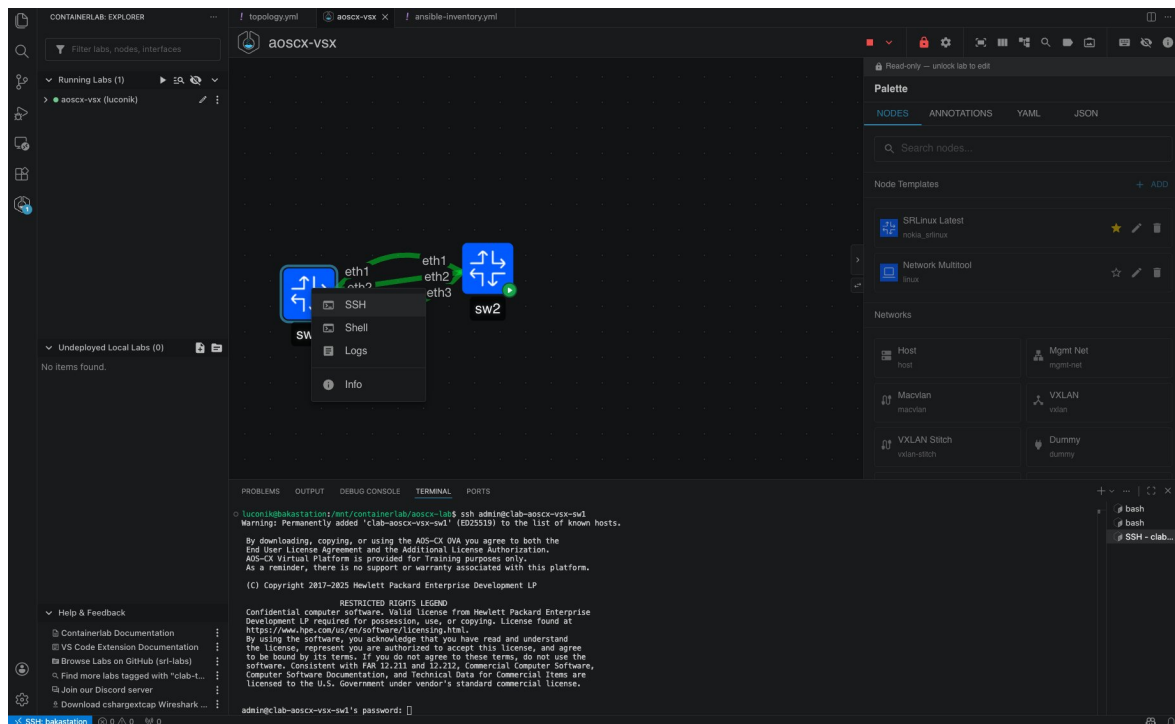
```
interface 1/1/2
  no shutdown
  lag 1
```

```
interface 1/1/3
  no shutdown
  ip address 192.168.255.0/31
```

VSX

```
system-mac 02:01:00:00:00:01
inter-switch-link lag 1
role primary
keepalive peer 192.168.255.1 source 192.168.255.0
```

■ "no routing" doit être configuré sur le LAG avant "inter-switch-link" / "no routing" must be set on the LAG before "inter-switch-link".



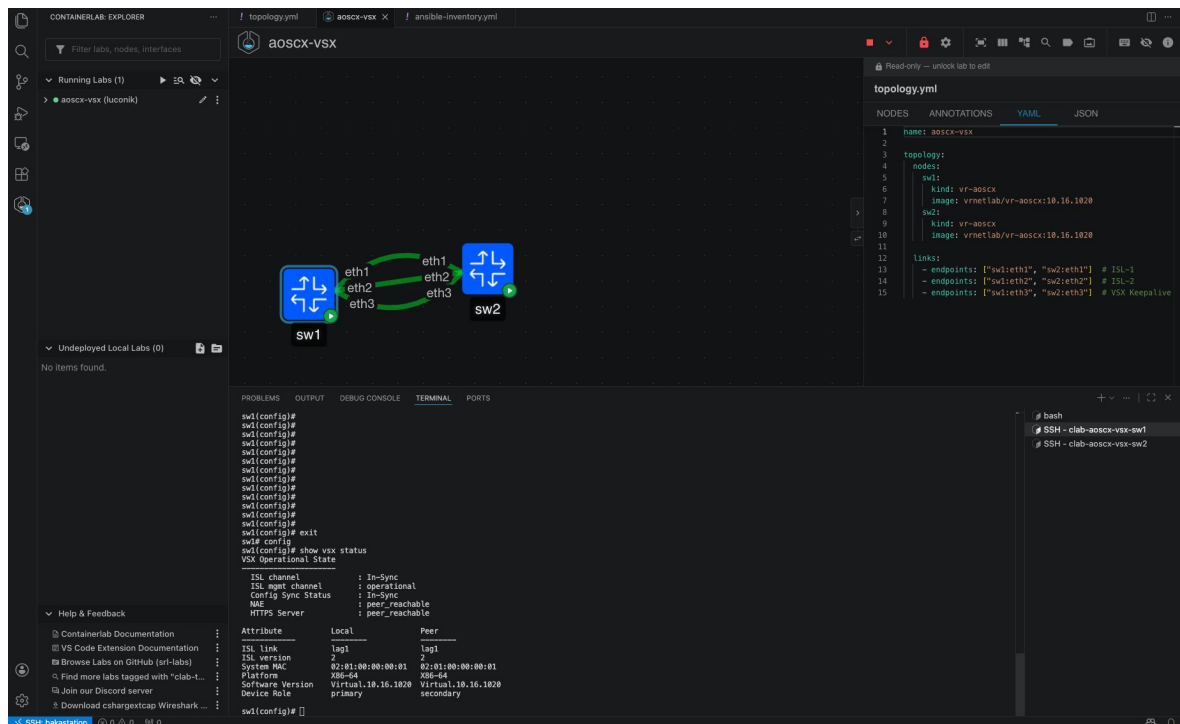
Plugin VS Code Containerlab — clic droit pour SSH direct sur un node VS Code Containerlab plugin — right-click for direct SSH on a node

```
configure terminal
```

```
vsx
system-mac 02:01:00:00:00:01
inter-switch-link lag 1
role secondary
keepalive peer 192.168.255.0 source 192.168.255.1
```

```
show vsx status
show lacp interfaces
show interface lag 1
```

Résultat attendu / Expected result — show vxr status : ISL channel: In-Sync | ISL mgmt channel: operational | Config Sync Status: In-Sync | NAE: peer_reachable



```
show vsx status — SW1 primary : ISL In-Sync, NAE peer reachable
```




CONTAINERLAB EXPLORER
Filter labs, nodes, interfaces
Running Labs (1)
aoscx-vsx (luconk)
Undeployed Local Labs (0)
Help & Feedback
Containerlab Documentation
VS Code Extension Documentation
Browse Labs on GitHub (sl-labs)
Find more labs tagged with "clab-L..."
Join our Discord server
Download cihargextcap Wireshark ...

aoscx-vsx

Read-only — unlock lab to edit
topology.yml

```

1 name: aoscx-vsx
2
3 topology:
4   nodes:
5     sw1:
6       kind: vr-aoscx
7       image: vrrnetlab/vr-aoscx:18.16.1820
8     sw2:
9       kind: vr-aoscx
10      image: vrrnetlab/vr-aoscx:18.16.1820
11
12   links:
13     - endpoints: ["sw1:eth1", "sw2:eth1"] # ISL-1
14     - endpoints: ["sw1:eth2", "sw2:eth2"] # ISL-2
15     - endpoints: ["sw1:eth3", "sw2:eth3"] # VSX Keepalive

```

PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS

```

swl(config)# show interface lag 1

Aggregate lag1 is up
Admin state is up
Description :
MAC Address : 88:08:09:63:0c:87
Aggregated-interfaces : 1/1/1 1/1/2
Aggregation-key : 1
Aggregation mode : active
Speed : 2000 Mb/s
QoS trust none
VLAN Mode: native-untagged
Active VLANs: 1
Allowed VLAN List: all
L3 Counters: Rx Disabled, Tx Disabled

Statistic      RX      TX      Total
-----
Packets        13488    14453        0
Unicast         0         0         0
Multicast       0         0         0
Broadcast       0         0         0
Bytes          9352284  12568832        0
Jumps           0         0         0
Dropped         0         0         0
Pause Frames    0         0         0
Errors          0         0         0
CRC/PCS         0         0         0
Collisions      0         0         0
Runt            0         0         0
Giants          0         0         0

```

```

# bash
# SSH - clab-aoscx-vsx-sw1
# SSH - clab-aoscx-vsx-sw2

```

show interface lag 1 — Aggregate lag1 up, 2x1G LACP = 2000 Mb/s

4. Lab Juniper vJunos-switch MC-LAG

■ La configuration MC-AE complète (active/standby) nécessite du matériel physique ou EVE-NG. Ce lab couvre l'ICL LACP et l'ICCP keepalive — base de la stack MC-LAG. Full MC-AE config (active/standby) requires physical hardware or EVE-NG. This lab covers ICL LACP and ICCP keepalive — the MC-LAG stack foundation.

Liens / Links

Lien / Link	Interfaces	Rôle / Role
ICL-1	sw1:ge-0/0/0 ↔ sw2:ge-0/0/0	Inter-Chassis Link (ae0)
ICL-2	sw1:ge-0/0/1 ↔ sw2:ge-0/0/1	Inter-Chassis Link (ae0)
ICCP	sw1:ge-0/0/2 ↔ sw2:ge-0/0/2	Keepalive 192.168.255.0/31
Client1	sw1:ge-0/0/3 ↔ client1:eth1	Ubuntu test client
Client2	sw2:ge-0/0/3 ↔ client2:eth1	Ubuntu test client

topology.yml

```
name: vjunos-mclag

topology:
  nodes:
    sw1:
      kind: juniper_vjunosswitch
      image: vrnetlab/juniper_vjunos-switch:25.4R1.12
      startup-config: sw1-startup.cfg
    sw2:
      kind: juniper_vjunosswitch
      image: vrnetlab/juniper_vjunos-switch:25.4R1.12
      startup-config: sw2-startup.cfg
    client1:
      kind: linux
      image: ubuntu:24.04
    client2:
      kind: linux
      image: ubuntu:24.04

  links:
    - endpoints: ["sw1:ge-0/0/0", "sw2:ge-0/0/0"] # ICL-1
    - endpoints: ["sw1:ge-0/0/1", "sw2:ge-0/0/1"] # ICL-2
    - endpoints: ["sw1:ge-0/0/2", "sw2:ge-0/0/2"] # ICCP
    - endpoints: ["sw1:ge-0/0/3", "client1:eth1"]
    - endpoints: ["sw2:ge-0/0/3", "client2:eth1"]
```

Déploiement / Deployment

```
sudo containerlab deploy -t topology.yml
sudo containerlab inspect --all
# Attendre healthy (~10-15 min) / Wait for healthy (~10-15 min)
ssh admin@172.20.20.7 # sw1 - admin/Lab12345!
ssh admin@172.20.20.4 # sw2 - admin/Lab12345!
```

■ Si SSH ne répond pas après "healthy" / If SSH does not respond after "healthy" : docker exec -it clab-vjunos-mclag-sw1 telnet 0 5000 login: root → configure → delete chassis auto-image-upgrade → commit

CONTAINERLAB EXPLORER

Filter labs, nodes, interfaces

Running Labs (2)

aoscx-vsx (luconik)

vjunos-mclag (luconik)

Undeployed Local Labs (0)

No items found.

Help & Feedback

Containerlab Documentation

VS Code Extension Documentation

Browse Labs on GitHub (for labs)

Find more labs tagged with "clab-l..."

Join our Discord server

Download cshargextcap Wireshark ...

vjunos-mclag

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
luconik@bakastation:~/mnt/containerlab/juniper-lab/vjunos-mclag$ sudo containerlab deploy -t topology.yml
17:22:15 INFO Containerlab started version=0.74.3
17:22:15 INFO Parsing & checking topology file= topology.yml
17:22:15 INFO Pulling image img=docker.io/library/ubuntu:24.04
24.04: Pulling from library/ubuntu

```

Topology	Lab Name	Name	Kind/Image	State	IPv4/6 Address
../..aoscx-lab/topology.yml	aoscx-vsx	clab-aoscx-vsx-sw1	vr-aoscx vnetlab/vr-aoscx:10.16.1820	running (healthy)	172.28.20.3 3fff:172:20:20::3
		clab-aoscx-vsx-sw2	vr-aoscx vnetlab/vr-aoscx:10.16.1820	running (healthy)	172.28.20.2 3fff:172:20:20::2
topology.yml	vjunos-mclag	clab-vjunos-mclag-client1	linux ubuntu:24.04	running	172.28.20.5 3fff:172:20:20::5
		clab-vjunos-mclag-client2	linux ubuntu:24.04	running	172.28.20.6 3fff:172:20:20::6
		clab-vjunos-mclag-sw1	juniper_vjunosswitch vnetlab/juniper_vjunos-switch:25.4R1.12	running (healthy)	172.28.20.7 3fff:172:20:20::7
		clab-vjunos-mclag-sw2	juniper_vjunosswitch vnetlab/juniper_vjunos-switch:25.4R1.12	running (healthy)	172.28.20.4 3fff:172:20:20::4

topology.yml

NODES

ANNOTATIONS

YAML

JSON

```

1 name: vjunos-mclag
2
3 topology:
4   nodes:
5     sw1:
6       kind: juniper_vjunosswitch
7       image: vnetlab/juniper_vjunos-switch:25.4R1.12
8       startup-config: sw1-startup.cfg
9
10    sw2:
11      kind: juniper_vjunosswitch
12      image: vnetlab/juniper_vjunos-switch:25.4R1.12
13      startup-config: sw2-startup.cfg
14
15    client1:
16      kind: linux
17      image: ubuntu:24.04
18
19    client2:
20      kind: linux
21      image: ubuntu:24.04
22
23 links:

```

Lab vjunos-mclag déployé — 2 switches + 2 clients Ubuntu, topology graphique VS Code vjunos-mclag lab deployed — 2 switches + 2 Ubuntu clients, VS Code graphical topology

Configuration MC-LAG — SW1

```
configure

set chassis aggregated-devices ethernet device-count 2

set interfaces ge-0/0/0 ether-options 802.3ad ae0
set interfaces ge-0/0/1 ether-options 802.3ad ae0

set interfaces ae0 aggregated-ether-options lacp active
set interfaces ae0 aggregated-ether-options lacp system-id 00:01:00:00:00:01
set interfaces ae0 aggregated-ether-options lacp admin-key 1
set interfaces ae0 unit 0 family ethernet-switching interface-mode trunk
set interfaces ae0 unit 0 family ethernet-switching vlan members all

set interfaces ge-0/0/2 unit 0 family inet address 192.168.255.0/31

set protocols iccp local-ip-addr 192.168.255.0
set protocols iccp peer 192.168.255.1 redundancy-group-id-list 1
set protocols iccp peer 192.168.255.1 liveness-detection minimum-interval 1000
set protocols iccp peer 192.168.255.1 liveness-detection multiplier 3

commit
```

Configuration MC-LAG — SW2

```
configure

set chassis aggregated-devices ethernet device-count 2

set interfaces ge-0/0/0 ether-options 802.3ad ae0
set interfaces ge-0/0/1 ether-options 802.3ad ae0

set interfaces ae0 aggregated-ether-options lacp active
set interfaces ae0 aggregated-ether-options lacp system-id 00:01:00:00:00:01
set interfaces ae0 aggregated-ether-options lacp admin-key 1
set interfaces ae0 unit 0 family ethernet-switching interface-mode trunk
set interfaces ae0 unit 0 family ethernet-switching vlan members all

set interfaces ge-0/0/2 unit 0 family inet address 192.168.255.1/31

set protocols iccp local-ip-addr 192.168.255.1
set protocols iccp peer 192.168.255.0 redundancy-group-id-list 1
set protocols iccp peer 192.168.255.0 liveness-detection minimum-interval 1000
set protocols iccp peer 192.168.255.0 liveness-detection multiplier 3

commit
```

■ *commit sur Junos = wr mem sur AOS-CX (config active ET persistante). "commit confirmed 5" = rollback automatique après 5 min si non confirmé — très utile en production. commit on Junos = wr mem on AOS-CX (config active AND persistent). "commit confirmed 5" = automatic rollback after 5 min if not confirmed.*

Validation MC-LAG

```
show iccp
show lacp interfaces
ping 192.168.255.1 count 5
```

Résultat attendu / Expected result — show iccp : TCP Connection: Established | Liveness Detection: Up | Redundancy Group 1: Up

Running Labs (2)

- asics-vxw (vucork)
- vjunos-mclag (vucork)
- client1 Up 54 minutes
- client2 Up 54 minutes
- sw1 Up 54 minutes (healthy)
- sw2 Up 54 minutes (healthy)

Undeployed Local Labs (0)

No items found.

Help & Feedback

- Containerlab Documentation
- VS Code Extension Documentation
- Browse Labs on GitHub (url-labs)
- Find more labs tagged with "clab-1..."
- Join our Discord server
- Download csharpcap Wireshark ...

Topology Diagram:

```

graph LR
    sw1[sw1] ---|ge-0/0/0| sw2[sw2]
    sw1 ---|ge-0/0/1| sw2
    sw1 ---|ge-0/0/2| sw2
    sw1 ---|eth1| client1[client1]
    sw2 ---|eth1| client2[client2]
  
```

Terminal Output:

```

error: configuration check-out failed

[edit]
admin@sw1# delete interfaces ae0 aggregated-ether-options mc-ae
[edit]
admin@sw1# commit
commit complete

[edit]
admin@sw1# show l2cp
syntax error,
admin@sw1# exit
Exiting configuration mode

admin@sw1> show la
'la' is ambiguous.
Possible completions:
  l2cp      Show Link Aggregation Control Protocol information
  layer2-control  Layer2-control information
admin@sw1> show l2cp interfaces
Aggregated interface: ae0
LACP state:
  ge-0/0/0 Actor No Yes No No No Yes Fast Active
  ge-0/0/1 Partner No Yes No No No Yes Fast Active
  ge-0/0/2 Partner No Yes No No No Yes Fast Active
LACP protocol:
  ge-0/0/0 Receive State Transmit State Mux State
  ge-0/0/1 Defaulted Fast periodic Detached Detached
  
```

show l2cp interfaces — ae0 actif avec ge-0/0/0 et ge-0/0/1 ae0 active with ge-0/0/0 and ge-0/0/1

Running Labs (2)

- asics-vxw (vucork)
- vjunos-mclag (vucork)
- client1 Up 55 minutes
- client2 Up 55 minutes
- sw1 Up 55 minutes (healthy)
- sw2 Up 55 minutes (healthy)

Undeployed Local Labs (0)

No items found.

Help & Feedback

- Containerlab Documentation
- VS Code Extension Documentation
- Browse Labs on GitHub (url-labs)
- Find more labs tagged with "clab-1..."
- Join our Discord server
- Download csharpcap Wireshark ...

Topology Diagram:

```

graph LR
    sw1[sw1] ---|ge-0/0/0| sw2[sw2]
    sw1 ---|ge-0/0/1| sw2
    sw1 ---|ge-0/0/2| sw2
    sw1 ---|eth1| client1[client1]
    sw2 ---|eth1| client2[client2]
  
```

Terminal Output:

```

[edit]
admin@sw2# set protocols iccp local-ip-addr 192.168.255.1
[edit]
admin@sw2# set protocols iccp peer 192.168.255.0 redundancy-group-id-list 1
[edit]
admin@sw2# set protocols iccp peer 192.168.255.0 liveness-detection minimum-interval 1000
[edit]
admin@sw2# set protocols iccp peer 192.168.255.0 liveness-detection multiplier 3
[edit]
admin@sw2# commit
commit complete

[edit]
admin@sw2# exit
Exiting configuration mode

admin@sw2> ping 192.168.255.1 count 5
PING 192.168.255.1 (192.168.255.1): 56 data bytes
64 bytes from 192.168.255.1: icmp_seq=0 ttl=64 time=0.043 ms
64 bytes from 192.168.255.1: icmp_seq=1 ttl=64 time=0.022 ms
64 bytes from 192.168.255.1: icmp_seq=2 ttl=64 time=0.030 ms
64 bytes from 192.168.255.1: icmp_seq=3 ttl=64 time=0.023 ms
64 bytes from 192.168.255.1: icmp_seq=4 ttl=64 time=0.047 ms
--- 192.168.255.1 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max/stddev = 0.023/0.033/0.047/0.010 ms
admin@sw2>
  
```

Ping keepalive 192.168.255.0 → 192.168.255.1 — 5/5 paquets, 0% perte, ~0.033ms Keepalive ping — 5/5 packets, 0% loss, ~0.033ms RTT

The screenshot displays the Containerlab Explorer interface. On the left, the 'Running Labs (2)' section lists 'vjunos-mclag' and 'vjunos-mclag (luconik)'. The main area shows a network topology diagram with two switches (sw1, sw2) and two clients (client1, client2). The connections are labeled with interface names: sw1 ge-0/0/0, sw2 ge-0/0/0, client1 ge-0/0/0, client2 ge-0/0/0, and eth1. The terminal window at the bottom shows the output of the 'show iccp' command, indicating that the TCP connection is established, liveness detection is up, and redundancy group 1 is up. The topology.yaml file on the right defines the network configuration, including the switches and clients.

```
1 name: vjunos-mclag
2
3 topology:
4   nodes:
5     sw1:
6       kind: juniper_vjunosswitch
7       image: vrnetlab/juniper_vjunos-switch:25.4R1.12
8       startup-config: sw1-startup.cfg
9
10    sw2:
11      kind: juniper_vjunosswitch
12      image: vrnetlab/juniper_vjunos-switch:25.4R1.12
13      startup-config: sw2-startup.cfg
14
15    client1:
16      kind: linux
17      image: ubuntu:24.04
18
19    client2:
20      kind: linux
21      image: ubuntu:24.04
22
23 links:
24
```

```
admin@sw1>
admin@sw1>
admin@sw1>
admin@sw1>
admin@sw1>
admin@sw1>
admin@sw1>
admin@sw1> show iccp
Redundancy Group Information for peer 192.168.255.1
TCP Connection      : Established
Liveness Detection  : Up
Redundancy Group ID : 1
Status              : Up
Client Application: l2ald_iccpd_client
Redundancy Group IDs Joined: None
Client Application: mclag_cfgchkd
Redundancy Group IDs Joined: 0 1
Client Application: lacpd
Redundancy Group IDs Joined: None
admin@sw1>
```

show iccp — TCP Connection Established, Liveness Detection Up, Redundancy Group 1 Up

5. Comparaison VSX vs MC-LAG

Élément / Element	AOS-CX VSX	Juniper MC-LAG
Technologie	VSX	MC-LAG + ICCP
ISL/ICL	LAG LACP	ae0 (LAG LACP)
Keepalive	IP dédiée / Dedicated IP	ICCP sur ge-0/0/2
Sync protocol	VSX natif	ICCP
Rôles / Roles	primary / secondary	active / standby
Config	vsx { }	protocols iccp + mc-ae
Vérification / Check	show vsx status	show iccp
Credentials	admin/admin	admin/Lab12345!
Boot time	~2-3 min	~10-15 min

6. Plugin VS Code Containerlab

L'extension **Containerlab** (publisher : srl-labs) pour VS Code offre une interface graphique complète pour gérer les labs réseau directement depuis l'éditeur. Compatible avec **VS Code Remote SSH** — accès depuis Mac ou Windows vers Linux.

- Visualisation graphique de la topologie en temps réel
- Deploy / Destroy / Inspect sans quitter VS Code
- Accès SSH direct par clic droit sur un node (SSH, Shell, Logs, Info)
- Édition du topology.yml avec autocomplétion YAML
- Affichage simultané de plusieurs labs dans "Running Labs"

Installation

VS Code → Extensions → Rechercher "Containerlab" → Publisher: srl-labs → Install

7. Commandes de référence / Reference Commands

Containerlab — Général / General

```
# Deploy / Destroy
sudo containerlab deploy -t topology.yml
sudo containerlab destroy -t topology.yml --cleanup
sudo containerlab deploy -t topology.yml --reconfigure

# Inspection
sudo containerlab inspect --all
sudo containerlab graph -t topology.yml # Web UI topologie

# Export configs
ssh admin@<ip> "show running-config" > sw1-config.txt
```

AOS-CX VSX

```
show vsx status
show vsx statistics
show lacp interfaces
show interface lag 1
show vsx brief
```

Juniper MC-LAG / ICCP

```
show iccp
show iccp statistics
show lacp interfaces
show interfaces ae0
show interfaces terse
ping 192.168.255.1 count 5
```

Console série vJunos / vJunos Serial Console

```
docker exec -it clab-vjunos-mclag-sw1 telnet 0 5000
# login: root (pas de mot de passe / no password)
# cli → configure → delete chassis auto-image-upgrade → commit
```